

Problem Statement

Students and researchers often struggle to quickly find relevant information from lengthy documents such as textbooks, notes, and research papers. Traditional keyword search methods are inefficient and fail to provide context-aware answers. This project addresses this limitation by using an AI-driven semantic search and answer generation system.

Proposed Solution

The system processes large PDF files by first extracting the text content and dividing it into smaller, meaningful chunks. These chunks are then transformed into vector embeddings that capture their semantic meaning. The embeddings are stored in a vector database for efficient retrieval.

When a user enters a question, the system converts the query into a vector representation and performs similarity search to retrieve the most relevant text chunks from the database. These retrieved sections are then passed to a generative AI model, which produces a clear, accurate, and context-aware response based strictly on the provided document content.

Methodology

- PDF text extraction and preprocessing
- Text chunking for efficient processing
- Conversion of text chunks into embeddings
- Storage in a vector database
- Query embedding and similarity-based retrieval
- Response generation using retrieved context (RAG pipeline)

Expected Outcome

The system will deliver precise, context-based answers directly from the uploaded PDF documents. This ensures high reliability, reduced search time, and improved understanding of complex materials.

Applications

- Academic study and exam preparation
- Research paper analysis
- Quick revision of textbooks and notes
- Digital learning assistance tools

Conclusion

The proposed AI-based Question Answering System using RAG significantly enhances information retrieval from large documents. By combining vector search with generative AI, it provides an intelligent and efficient solution that improves student productivity and learning experience.